

2022-25(Voc.)

Time : 3 hours

Full Marks : 70

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Answer from both the Groups as directed.

Group – A

Answer any **four** questions of the following :

10×4 = 40

1. Define TCP / IP model in details. Discuss its layers and functions. Explain different protocols used at different layers with suitable examples.
2. Explain how an internet server establishes connectivity and verify its identity on the internet.
3. Define array and explain their types with example in JavaScript.

4. What are conditional statements in JS ? Explain their types with suitable examples.
5. Explain the term XML and HTML and how HTML differ from XML. Also write an HTML program to display an image.
6. (a) Discuss the roles of SGML and XML for creating dynamic web pages.
(b) Explain how XML data are modeled.
7. What are ISP directives ? Name and explain its different types.
8. What do you mean by PHP variables and datatypes ? Explain with examples.

Group – B

Answer all questions :

3×10 = 30

9. What are IP addresses ?
10. What are the differences between Web servers and Web clients ?
11. Define data types used in JavaScript.

- ✓ 12. Explain the advantages of JavaScript for creating a webpage.
13. Explain the steps to apply different styles of XML with XSL.
- ✓ 14. Explain the term 'Dynamic Web Pages'.
15. What is scripting in JSP ? Explain. *Right*
- ✓ 16. Explain the different types of operators used in JavaScript with examples.
17. Explain any six string functions used in PHP.
18. Write some advantages of Internet.



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Section – A

Answer any four questions of the following :

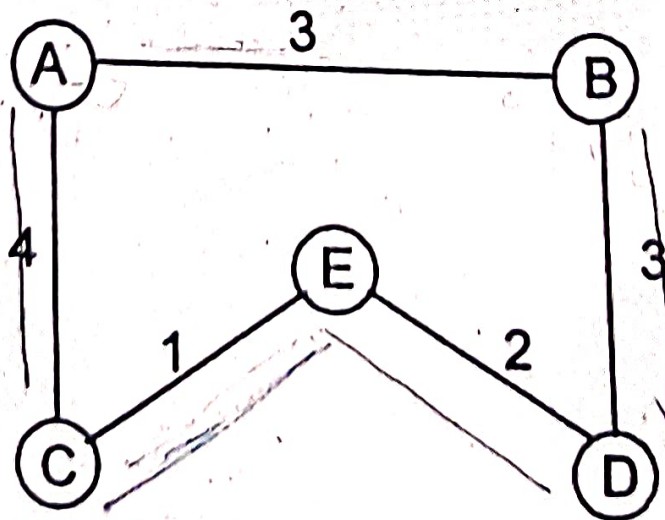
10×4 = 40

1. (a) Define an algorithm. What are its characteristics ?

(b) What is asymptotic notation ? Explain with an example.

2. Explain the binary search algorithm with an example.

3. (a) Define sorting. What are some common sorting techniques ?
- (b) Write an algorithm for merge sort.
4. What is collision in hashing ? Explain different methods to handle collisions.
5. (a) What is a spanning tree ? Give an example.
- (b) Define a minimum spanning tree (MST) and its applications.
6. Explain Kruskal's algorithm with a step by step example. Apply Kruskal's algorithm to find minimum cost spanning tree for the graph given below :



7. Differentiate between Binary tree and Binary search tree. Write an algorithm to construct a binary search tree. Also construct a BST for following nodes :

10, 8, 15, 7, 3, 6, 12, 5, 9, 17.

8. Define Graph. Explain BFS and DFS traversing technique in graph.

Section – B

Answer all questions of the following : $3 \times 10 = 30$

9. Differentiate between directed and undirected graph.

10. Write a recursive algorithm to find factorial of a given number.

11. Write an algorithm to find largest number among three given numbers.

12. Sort following using merge sort show steps :
38, 27, 43, 12, 25, 60.

13. Differentiate between BST and AVL tree.

14. Define weighted and unweighted graph.

15. Define data structure. List common types of data structure.

16. Write properties of Binary search tree.

17. Define recursive function. What is the base case in recursion ?

18. Define time complexity and space complexity.



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Section – A

Answer any four questions of the following :

10×4 = 40

1. What is the Linux file system ? Explain the key components.
2. Briefly explain about Linux operating system and their features.
3. Explain the use of 'grep' and 'sed' commands in Linux with example.

4. (a) What are environmental variable in shell scripting ? Explain with example.
(b) What is an if else statement in shell scripting ? Explain with example.
5. Explain Loop in shell scripting. Explain all types of Loop with example.
6. What are arithmetic and string operators in AWK ? Explain with example.
7. What is a system call in Linux ? Explain its type with example.
8. What is the standard I / O package in Linux ? Explain its significance.

Section – B

Answer all questions of the following : $3 \times 10 = 30$

9. What is the role of Kernel in Linux ?
10. What is the function of VI editor ?
11. Define a shell script.
12. Use of begin and end block in AWK.

13. What is file locking in Linux ?

14. What are file permissions in Linux ?

15. What is a pipeline in Linux ?

16. Explain 3 commands in Linux OS.

17. What is an alias in Linux ? Explain with example.

18. What is an AWK built-in variable ?



Environ

Divided by 10

Real root

RK

Gauss dimension

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Group – A

Answer any **four** questions of the following :

10×4 = 40

1. Apply Newton Raphson method to find roots of equation $x^3 - 2x - 5 = 0$.

2. Define bisection method. Write steps to find root of equation using bisection method.

Also find a real root of :

$f(x) = x^3 - 4x - 9 = 0$ using bisection method.

3. (a) Derive Simpson's 1 / 3 rule for numerical integration.

(b) Compute value of $\int_0^1 \frac{dx}{1+x}$ using Simpson's

1 / 3 rule with $h = \frac{1}{6}$.

4. Derive trapezoidal rule for numerical integration.

Evaluate $\int_0^1 \frac{dx}{1+x^2}$ using trapezoidal rule with

$h = 0.2$.

5. (a) Derive Newton forward difference interpolation formula.

(b) Evaluate $\left(\frac{\Delta^2}{E} \right) x^3$.

6. Using Lagrange's interpolation formula find $f(7)$ for the given data points :

x	$f(x)$
2	5
4	9

x	$f(x)$
6	15
8	25

7. (a) Discuss absolute error, relative error and percentage error with example.

(b) Explain general error formula.

8. Solve the equations using Gauss-Jordan method :

$$3x + 2y - z = 5$$

$$x - y + 2z = 3$$

$$2x + y + z = 4$$

Group – B

Answer all questions :

$3 \times 10 = 30$

9. Differentiate between interpolation and extrapolation.

10. What is numerical differentiation ?

11. Where is Simpson's rule used ?

12. List the types of Runge-Kutta method.

13. What is Gauss-Jacobi method ?

14. Show relation between Δ , ∇ and E .

15. Compare Gauss elimination and Gauss-Jordan method.

16. Write three reasons due to which an error can occur.

17. What is rounding rule?

18. What is shift operator?



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Section – A

Answer any four questions of the following :

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1. Explain testing and debugging. Also write differences between testing and debugging.

2. What are the purpose of software testing ? Explain.

3. Explain path products and path expression with example and also write their application in software testing .

4. What are decision tables ? Explain in details.
5. Explain different level of testing in s /w testing.
6. How can software be represented as a graph for testing purpose ? Write common application of graph based testing.
7. What is domain testing ? Explain and write their advantage.
8. What are the flow testing techniques ? Discuss in details.

Section – B

9. Answer all questions of the following :

3×10 = 30

(a) What are bugs ? Name their types.

(b) Define blackbox testing and whitebox testing.

(c) What is Vasisification ?

(d) What are metrics and their types ?

(e) What is a test case ?

(f) Explain testing approaches in brief.

(g) What is software testing ? Name their two main types.

(h) Explain bug life cycle with diagram.

(i) What is a path testing ?

(j) What is software quality ? What are the need of software quality ?

